

Telematics & Gateway ECU v3.0 Technical Specification

1. Introduction

The Telematics & Gateway ECU v3.0 is a next-generation automotive electronic control unit designed for secure vehicle telematics, network bridging, and gateway management. This document outlines the electrical, thermal, and interface requirements for the component.

2. Operating Voltage

The ECU requires a nominal 12V DC power supply from the vehicle power net. The device includes power conditioning and transient protection to handle voltage drops down to 9V while maintaining normal operation. Operating below 9V is not recommended and may cause reduced functionality or system resets.

3. Current Draw

The maximum current draw for the Telematics & Gateway ECU v3.0 is 15 Amps under peak operating conditions. Normal steady-state current draw is expected to be below 5 Amps during active operation, with additional current demand during startup and high-load communication events. The power architecture should account for transient surge capability and circuit protection at 15 Amps maximum.

4. CAN Bus Interfaces

The ECU supports multiple CAN bus interfaces to enable vehicle network bridging, diagnostics, and telematics data exchange. Interfaces include dual high-speed CAN channels compliant with ISO 11898-2 and one medium-speed CAN channel for body control communication. Each CAN port includes galvanic isolation and configurable termination to support line topology and noise resilience.

5. Operating Temperature

The operating temperature range for the ECU is -40°C to 85°C. The design uses automotive-grade components, thermal management, and conformal coating to ensure reliable performance across this range. Elevation beyond 85°C may result in thermal throttling or degraded component life.

6. Sleep Mode Power Consumption

In Sleep Mode, the Telematics & Gateway ECU v3.0 reduces power consumption by disabling non-critical subsystems while maintaining wake-up capability for network events. Sleep current is targeted at under 5 mA from the 12V supply, including all retention logic and wake event monitoring circuits. This enables minimal battery drain during parked vehicle operation.

7. Mechanical and Environmental Considerations

The enclosure is designed for automotive environments with ingress protection for in-cabin or under-dash mounting. Vibration testing follows ISO 16750 standards, and the module includes EMI filtering for reliable operation in harsh electromagnetic environments.

8. Summary

The Telematics & Gateway ECU v3.0 provides robust power handling, comprehensive CAN connectivity, and automotive temperature tolerance. Design verification should focus on the 12V nominal supply, 9V drop tolerance, 15 Amp maximum current capability, and low sleep mode power consumption.